WACHEMO UNIVERSITY
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF SOFTWARE ENGINEERING

SEng7232 – Fundamentals of Cloud Computing

Year III, Semester II | Academic Year 2025/2026

THEORETICAL ASSIGNMENT

Cloud Products, Services, Operations & Management in Cloud Computing

Course Code: SEng7232

Assignment Weight: 15% of the final course grade

Total Marks: 100 marks

Chapters Covered: Chapter 3 – Cloud Products, Services, and Platforms
Chapter 6 – Cloud Operations and Management

Issue Date: ____________________

Submission Deadline: ____________________

Student Information			
Full Name:		**ID No:**	
Section / Group:		**Signature:**	

Assignment Instructions

General Instructions

1. This is an **individual assignment**. All submitted work must be your own. Collaboration, copying, or any form of academic dishonesty will result in a mark of zero.
2. The assignment consists of **four parts**. Read each question carefully before writing your response.
3. Answer **all questions** in all parts. No optional questions are provided.
4. Responses must be written in **formal, academic English**. Answers written in point form only, without adequate explanation, will not receive full marks.
5. Structure your answers clearly using paragraphs, labelled diagrams (where applicable), and organised comparisons.
6. The **total marks** for this assignment is **100 marks**, corresponding to **15%** of the final course grade.
7. Marks for each question are indicated in square brackets [ ].
8. This assignment covers **Chapter 3** (Cloud Products, Services, and Platforms) and **Chapter 6** (Cloud Operations and Management).
9. **Handwritten** submissions must be legible, written in black or dark-blue ink. **Typed** submissions must use Times New Roman or Arial, 12 pt, 1.5 line spacing.
10. Late submissions will be penalised at **10% per calendar day** beyond the deadline, up to a maximum of three days. Submissions received more than three calendar days late will receive a mark of zero.

Important Notice

Students are expected to draw on knowledge developed in lectures, course materials, and recommended textbooks. Direct quotation of definitions without further explanation or application will not demonstrate sufficient understanding and will be marked accordingly. All written responses should reflect critical thinking and the ability to apply concepts to realistic scenarios.

Mark Distribution Summary

Part	Topic Area	Questions	Marks
Part A	Cloud Products, Services, and Platforms Concepts and Comparison	Q1–Q3	30
Part B	Cloud Networking, Storage, and Database Services	Q4–Q5	25
Part C	Cloud Operations, Cost, and Resource Management	Q6–Q8	30
Part D	Integrated Analysis and Critical Evaluation	Q9	15
Total		**9 Questions**	**100**

Part A – Cloud Products, Services, and Platforms: Concepts and Comparison [30 Marks]

Question 1 12 Marks

Cloud Service Models and Provider Ecosystem

a. Explain the conceptual distinction between **Infrastructure-as-a-Service (IaaS)** and **Platform-as-a-Service (PaaS)** with respect to the *division of operational responsibility* between the cloud provider and the customer. Identify at least **four** specific responsibilities and indicate which party holds each under both models. [**5 marks**]

b. The three major cloud providers – AWS, Microsoft Azure, and GCP – offer **functionally equivalent services** under different product names. Using the service categories of *compute*, *object storage*, *block storage*, *managed database*, and *PaaS application hosting*, identify the corresponding service name from each provider and briefly explain why such conceptual equivalence is significant for cloud architects. [**5 marks**]

c. NIST defines five essential characteristics of cloud computing. Select **three** of those characteristics and, for each, identify a specific cloud service category from Chapter 3 that directly realises it. Justify each pairing with a concise explanation. [**2 marks**]

Question 2 10 Marks

Virtual Machine Instances and Computing Services

a. Define the following core IaaS terms and explain the functional role of each within a cloud environment: **(i)** Instance Type, **(ii)** Amazon Machine Image (AMI) or equivalent OS snapshot, **(iii)** Availability Zone (AZ), and **(iv)** Auto Scaling. [**4 marks**]

b. A student claims: *"Choosing a larger instance type always results in better application performance and is therefore always the correct design decision."* Critically evaluate this claim. Discuss the relationship between instance sizing, cost, actual utilisation, and the cloud principle of *measured service*. [**3 marks**]

c. Describe the sequence of steps required to provision a virtual machine instance on a major cloud platform of your choice. For each step, briefly explain the purpose of the configuration decision being made. [**3 marks**]

Question 3 8 Marks

IaaS versus PaaS: Service Selection and Trade-offs

a. For each scenario below, identify whether **IaaS** or **PaaS** is more appropriate and provide a justification of no more than three sentences:

i. A research group requires a custom Fortran-based HPC simulation that depends on specific OS kernel parameters and low-level compiler flags. [**1.5 marks**]

ii. A four-person final-year project team is building a standard Django web application with a ten-week deadline and no dedicated systems administration expertise. [**1.5 marks**]

iii. A hospital IT department is deploying a vendor-supplied, closed-source Windows Server application that requires a specific OS version and cannot be modified. [**1.5 marks**]

b. State the **core design principle** that should guide a cloud engineer when choosing between IaaS and PaaS, and explain why adhering to this principle tends to reduce total cost of ownership and operational risk. [**3.5 marks**]

Part B – Cloud Networking, Storage, and Database Services [25 Marks]

Question 4 13 Marks

Storage Services and Network Architecture

a. Compare and contrast **object storage** and **block storage** across the following dimensions: access method, scalability, latency characteristics, attachment model, and best-fit use cases. Present your comparison as a **structured written discussion**, explaining the technical reason behind each difference rather than merely listing facts. **[5 marks]**

b. A development team is architecting a new social media platform requiring: **(i)** storage for user-uploaded profile images and video clips accessible from any location via a web URL; **(ii)** storage for the root OS volume of the application servers; and **(iii)** storage for the PostgreSQL database data directory requiring sub-millisecond I/O latency. For each requirement, specify the appropriate storage type, name the corresponding service on one cloud provider, and justify your selection. **[4 marks]**

c. Explain the purpose and architecture of a **Virtual Private Cloud (VPC)**. Define and explain the role of each of the following VPC components: subnet (public and private), Internet Gateway, Security Group, and NAT Gateway. **[4 marks]**

Question 5 12 Marks

Managed Database Services

a. Define a **managed database service** and explain how it differs from deploying a database engine on a self-managed virtual machine. Compare the two approaches across: OS patching, backup management, replication, automated failover, and high availability configuration. **[5 marks]**

b. Critically assess the following statement:

> *"Self-managed databases on virtual machine instances are preferable to managed database services because they provide greater control, are cheaper per month, and are more flexible for all production workloads."*

Identify the conditions under which each approach may be justified and explain the *hidden costs* of self-managed database operation not reflected in the compute instance price alone. **[4 marks]**

c. Identify and briefly describe **three distinct categories** of managed database services available on modern cloud platforms (e.g., relational SQL, serverless SQL, NoSQL, in-memory cache). For each category, name one representative service from any major provider and state a scenario in which it would be preferred over the others. **[3 marks]**

Part C — Cloud Operations, Cost, and Resource Management
[30 Marks]

Question 6 10 Marks

Cloud Resource Monitoring and Management

a. Describe **four ways** in which a software engineer can interact with and manage cloud resources, from the most manual to the most automated approach. For each method, explain its primary use case, its key advantage, and one significant limitation. [**4 marks**]

b. Cloud management consoles provide a graphical interface for resource lifecycle management. Outline the **general lifecycle stages** a cloud resource passes through from initial provisioning to termination, and for each stage explain the operational significance and any associated risks. [**3 marks**]

c. Explain the concept of **resource monitoring** in cloud environments. Discuss why monitoring metrics such as CPU utilisation, storage consumption, and network throughput are essential for both operational reliability and cost control, and describe the consequences of deploying resources without adequate monitoring. [**3 marks**]

Question 7 11 Marks

Cloud Costs, Billing, and Scaling

a. Explain the **pay-as-you-go pricing model** that underpins public cloud billing. Describe how this model relates to the NIST characteristic of *Measured Service* and explain how it differs fundamentally from the capital expenditure model of traditional on-premises data centres. [**3 marks**]

b. Discuss **five concrete strategies** that a software engineering team can apply to reduce unnecessary cloud expenditure without compromising availability or performance. For each strategy, explain the mechanism by which it reduces cost and identify the type of workload to which it is most applicable. [**5 marks**]

c. Distinguish between **manual scaling** and **automatic scaling** in cloud environments. Explain the conditions under which each approach is appropriate, and describe the potential operational and financial consequences of failing to scale resources in response to traffic demand changes. [**3 marks**]

Question 8 9 Marks

Cloud Reliability, Availability, and Future Trends

a. Define **uptime** and **availability** in the context of cloud computing. Explain how Multi-Availability-Zone (Multi-AZ) deployments, load balancing, and automated failover collectively contribute to achieving high availability for a cloud-hosted application. [**4 marks**]

b. Describe the principles of **data backup and disaster recovery** in cloud environments. Define the terms *Recovery Point Objective (RPO)* and *Recovery Time Objective (RTO)*, and explain how managed cloud services address each compared to manually managed backup procedures on virtual machines. [**3 marks**]

c. Identify and briefly describe **two emerging trends** in cloud computing likely to influence cloud-based software systems over the next five years. Discuss the potential implications of each trend for software engineering professionals. [**2 marks**]

Part D – Integrated Analysis and Critical Evaluation [15 Marks]

Question 9 15 Marks

Cloud Architecture Design and Evaluation – Case Analysis
Read the following scenario carefully, then answer all sub-questions.

Scenario: Regional E-Government Portal – Haddiya Zone Digital Services

The Haddiya Zone Administration plans to launch a **citizen services web portal** allowing residents to apply for identification documents, pay utility bills, and access public records online. System requirements:

- Expected concurrent users: up to **8,000** during peak periods (typically month-end and public holidays); near-zero traffic at other times.
- Application stack: a **Python-based web framework** with a **PostgreSQL** relational database for citizen records.
- Storage: citizen documents (scanned PDFs and photographs) must be stored securely for a **minimum of seven years** to comply with government record-keeping regulations.
- The system must maintain **99.9% availability** and recover from any database failure within five minutes.
- The system will be operated by a small team of **four software engineers** with limited cloud infrastructure experience.
- A preliminary annual cloud budget of **USD 6,000** has been approved.

a. Service Selection and Justification. For each of the four core service pillars (compute, storage, networking, and database), identify the *specific cloud service and model* (IaaS or PaaS) you would recommend. For each recommendation, provide a written justification referencing at least one specific requirement stated in the scenario. [**5 marks**]

b. Availability and Reliability Architecture. Explain how you would design the cloud infrastructure to meet the **99.9% availability** requirement and the **five-minute database recovery** target. Describe the specific architectural mechanisms (e.g., multi-AZ deployment, load balancing, automated failover) that achieve these guarantees, and explain the purpose of each. [**4 marks**]

c. Cost and Scalability Analysis. Given the highly variable traffic pattern (near-zero for most of the month with brief peaks), analyse how the choice of service model and scaling strategy influences *cost efficiency*. Identify which NIST cloud characteristic most directly enables cost savings between peak periods, and explain how this characteristic is realised by the services you recommended. [**3 marks**]

d. Risk Identification. Identify **two** architectural or operational risks associated with the scenario, and for each risk propose a specific mitigation strategy grounded in the cloud operations and management principles studied in Chapter 6. [**3 marks**]

Submission Requirements

Submission Guidelines

1. **Format:** Submissions may be *handwritten* (legible, black or dark-blue ink) or *typed* (MS Word or PDF; Times New Roman or Arial, 12 pt, 1.5 line spacing, minimum 2.5 cm margins).
2. **Cover Page:** All submissions must include a completed cover page bearing the student's full name, ID number, section, and a signed declaration of academic integrity.
3. **Page Guideline:** There is no strict page maximum, but responses totalling 8–14 pages (excluding cover page) are expected for full-mark answers. Unnecessary repetition will not attract additional marks.
4. **Diagrams:** Hand-drawn or digital diagrams are acceptable where included. Each diagram must be clearly labelled and explicitly referenced within the written response.
5. **References:** If external sources beyond the course materials are consulted, cite them using APA 7th edition format in a reference list appended to the submission.
6. **Submission Method:** Submit a *physical hard copy* to the Department of Software Engineering administrative office **or** upload a PDF via Google Classroom code **7bhxgumf** before the stated deadline.
7. **File Naming (digital submissions):** `SEng7232_Assign_[StudentID]_[Surname].pdf`
8. **Academic Integrity Declaration:** By submitting this assignment, the student affirms that the work is entirely their own, produced without unauthorised assistance, and has not been submitted for assessment in any other course.

Detailed Marking Rubric

The following rubric defines the criteria and performance levels used to evaluate all responses. Markers apply these descriptors holistically across all questions unless question-specific criteria are listed below.

General Performance-Level Descriptors

Level	Grade Range	Descriptor
Distinction	**90–100%**	Thorough, accurate, and detailed understanding of all concepts. Fluent application to novel and complex scenarios. Logically structured, analytically rigorous, and precise. All key terms correctly defined and contextualised. Evidence of independent critical thinking and synthesis across concepts.
Credit	**75–89%**	Sound understanding of most concepts with minor omissions. Competent application with generally correct justification. Well organised and mostly complete. Minor errors in terminology that do not affect overall correctness.
Pass	**50–74%**	Basic understanding of core concepts but with notable gaps or errors. Partial application with limited justification. Responses may be poorly structured or insufficiently detailed. Some key terms may be misused or absent.
Fail	**0–49%**	Insufficient or incorrect understanding. Responses largely irrelevant, incoherent, or incomplete. Key terms absent, misused, or contradictory. Little or no meaningful application to the given question or scenario.

Question-Specific Marking Criteria

Question	Marks	Key Marking Criteria	Full Marks	Partial Marks
Q1a	5	Clear IaaS vs. PaaS responsibility delineation. Correct identification of $\geq$4 responsibilities, each accurately assigned to provider or customer under both models.	All 4+ responsibilities correctly assigned with explanation.	1 mark per correctly explained responsibility (max 4); 1 mark for overall clarity.
Q1b	5	Correct service names across all five categories for all three providers. Meaningful explanation of why conceptual equivalence matters for portability.	All 15 service-provider pairings correct plus explanation.	0.5 marks per correct pairing (max 4); 1 mark for explanation.

Question	Marks	Key Marking Criteria	Full Marks	Partial Marks
Q1c	2	Three NIST characteristics correctly identified and mapped to a service category with valid justification for each pairing.	All three pairings correct with valid justification.	0.5 marks per correctly justified pairing.
Q2a	4	All four terms (Instance Type, AMI/OS snapshot, AZ, Auto Scaling) accurately defined. Functional role explained in operational context.	All four terms defined accurately with role explained.	0.5 marks for definition + 0.5 marks for role per term.
Q2b	3	Identifies that larger instances increase cost without proportional performance gain if underutilised. Links to measured service and right-sizing.	Counter-argument substantiated with ≥2 concrete reasons linked to cost and measured service.	1 mark per substantiated point (max 3).
Q2c	3	Correct logical sequence of VM provisioning steps. Purpose of each step explained (AMI, instance type, VPC/SG, storage, key pair).	≥5 steps in logical order with purpose explained for each.	0.5 marks per step with purpose explained.
Q3a	4.5	For each of three scenarios: correct IaaS/PaaS selection; justification references the specific technical or operational reason. Vague justifications receive half marks.	All three correct with clear, specific justification.	0.5 marks for correct choice + 1 mark for justification per scenario.
Q3b	3.5	Core principle stated clearly (highest level of abstraction satisfying requirements). Explanation links to TCO reduction and reduced operational risk with specific reasoning.	Principle stated correctly + explanation of TCO and risk with ≥2 specific reasons.	1 mark for principle; up to 2.5 marks for explanation quality.
Q4a	5	Structured discussion addressing all five dimensions. Technical *reasons* for each difference provided, not merely facts listed. Discussion is coherent and comparative throughout.	All five dimensions covered with technical explanation.	1 mark per dimension fully addressed (definition + reason + comparison).

Question	Marks	Key Marking Criteria	Full Marks	Partial Marks
Q4b	4	For each of three requirements: correct storage type identified; correct specific service named on one provider; justification references a technical property matching the requirement.	All three requirements correctly answered with valid justification.	0.5 marks for type + 0.5 marks for service + 0.33 marks for justification per requirement.
Q4c	4	VPC purpose and architecture clearly explained. All five components (subnet public/private, Internet Gateway, Security Group, NAT Gateway) defined correctly with role explained.	VPC overview and all five components correctly explained.	0.5 marks for VPC overview + 0.7 marks per component.
Q5a	5	Managed database service defined. Comparison covers all five responsibilities (patching, backup, replication, failover, HA) for both self-managed and managed models.	All five responsibilities compared correctly across both models.	0.5 marks per responsibility correctly compared across both models.
Q5b	4	Engages critically with the statement. Identifies conditions where self-managed is justified. Discusses hidden costs (engineering hours, outage risk, manual patching) not reflected in compute price.	Both sides evaluated; $\geq$3 hidden costs identified with explanation.	1 mark for valid self-managed cases; 3 marks for hidden cost discussion.
Q5c	3	Three distinct database categories correctly identified. One representative service per category named. Preferred scenario for each is plausible and clearly justified.	All three categories correct with service named and scenario justified.	1 mark per category (0.3 + 0.3 + 0.4).
Q6a	4	All four interaction methods (Console, CLI, SDK, IaC) described. For each: primary use case, key advantage, and limitation stated. Ordered from most manual to most automated.	All four methods with use case, advantage, and limitation.	1 mark per method fully described.
Q6b	3	Lifecycle stages outlined logically (create, configure, monitor, terminate). Operational significance and at least one risk per stage identified.	All major stages with significance and risk.	0.75 marks per stage fully discussed.

Question	Marks	Key Marking Criteria	Full Marks	Partial Marks
Q6c	3	Monitoring defined clearly. Link between key metrics and operational reliability explained. Consequences of absent monitoring discussed with $\geq$2 specific negative outcomes.	Monitoring concept + metric relevance + $\geq$2 consequences.	1 mark each for: definition, metric relevance, consequence discussion.
Q7a	3	Pay-as-you-go model correctly explained. Correct linkage to NIST Measured Service. Clear distinction from CapEx model with at least two differences.	All three elements addressed correctly.	1 mark per element.
Q7b	5	Five strategies identified (e.g., right-sizing, reserved instances, auto scaling, shutting idle resources, storage lifecycle policies). For each: mechanism of cost reduction and applicable workload type explained.	All five strategies with mechanism and workload type.	1 mark per strategy fully described.
Q7c	3	Manual and auto scaling clearly distinguished. Conditions for each identified. Consequences of failure to scale (cost waste and service degradation) discussed.	Both types distinguished with conditions and consequences.	1 mark per element.
Q8a	4	Uptime and availability defined correctly. Multi-AZ, load balancing, and automated failover each explained with their collective contribution to HA described.	Both definitions + all three mechanisms with contribution explained.	0.5 marks per definition; 1 mark per mechanism.
Q8b	3	RPO and RTO correctly defined. How managed services address each (automated backups, Multi-AZ failover) explained and contrasted with manual procedures.	Both terms defined; both addressed by managed services; contrast drawn.	0.5 marks per definition; 1 mark per RPO/RTO explanation; 0.5 marks for contrast.
Q8c	2	Two distinct, plausible emerging trends identified (e.g., serverless, edge computing, AI/ML integration, green cloud). Implications for software engineers discussed with at least one concrete impact each.	Both trends + specific SE implication per trend.	0.5 marks for each trend + 0.5 marks for each implication.

Question	Marks	Key Marking Criteria	Full Marks	Partial Marks
Q9a	5	All four pillars addressed. Service and model (IaaS/PaaS) recommended for each. Each recommendation references a specific stated scenario requirement. Recommendations are mutually consistent.	All four pillars with service, model, and scenario-referenced justification.	1 mark per pillar fully justified.
Q9b	4	Multi-AZ deployment, load balancing, and automated failover described and linked to the 99.9% and 5-minute targets. Purpose of each mechanism explained.	All three mechanisms described and linked to both stated targets.	Up to 1.3 marks per mechanism with target linkage.
Q9c	3	Variable traffic pattern and cost implications analysed. Correct NIST characteristic identified (Measured Service). Linkage between recommended services and cost efficiency during low-traffic periods explained.	Analysis of traffic pattern + correct NIST characteristic + linkage to recommendations.	1 mark per element.
Q9d	3	Two distinct architectural/operational risks identified. Each risk is specific to the scenario. Each mitigation is grounded in a Chapter 6 principle and is actionable.	Both risks scenario-specific; both mitigations Chapter 6-grounded.	0.5 marks per risk identified + 1 mark per mitigation.

Important Notice

Additional Marking Notes for Assessors:

- Marks are not awarded for bullet-point answers that lack explanation. Students must demonstrate *understanding*, not merely recall.
- Correct identification of a service name without explanation of its functional role earns a maximum of **half marks** for that element.
- Diagrams that are relevant and correctly labelled may supplement a written answer but do not substitute for written explanation.
- Penalty for academic integrity violations: as per WCU Academic Regulation — minimum penalty is zero for the entire submission.
- Late penalty: **10% per calendar day**; zero after 3 days.